

Capstone Project

Financial Growth Audit for a B2B SaaS Company

Executive Summary

- **Strong Growth with Hidden Profit Leaks:** The business achieved impressive growth since 2021, with monthly recurring revenue increasing by 68% and monthly profits surging by 142%. However, excessive discounting has damaged the bottom line, with discounts over 20% causing a total loss of \$135.3k.
- **Perfect Customer Retention but No New Customers:** The company has an exceptional 100% customer retention rate, and a net retention rate of over 1600%, meaning no clients have left, and existing accounts have expanded significantly. However, for over 3 years no new customers were being acquired, which impacted the potential growth over the years.
- **Business is ready for expansion.** An average customer lifetime value of \$42k proves the business has the financial health to safely spend between \$8.4k and \$14k to acquire new accounts.
- **Stable Customer Base and Future Capital:** The current customer base is highly secure and predictable, with 55 out of 99 active enterprise clients classified as top-tier loyal accounts and only 4 accounts flagged as at-risk.
- **Product Profit Issues:** Marketing Suite is the only unprofitable product line; because its discounts are normal, its losses are likely caused by other factors that need to be investigated.

Introduction

In this project, I'm using transaction data by a fictional SaaS (software-as-a-service) company to derive insights, regarding the company's overall financial health and opportunities for further growth.

This company provides sales and marketing software to other companies (B2B), which come from a variety of industries, such as finance, energy, tech, manufacturing,

healthcare, transportation, retail, among others. The customer base is also global, as they make purchases from different parts of the globe.

Data that I'm working with was collected in a 4-year period, from 2020 to 2023, which allows for exploration of multi-year trends regarding revenue growth, profitability, effectiveness of discounting strategies and the performance of customers across different segments.

Core business objective

- Where is **profit loss** occurring and how can we **leverage high-value customer segments** to drive further growth?

Plan of action

1. Profit loss - identify the causes and prepare solutions.
 - How do discounts affect profitability?
 - How does customer churn impact growth?
2. Identifying high-value customers
 - How does sales performance differ across cohorts?
 - How engaged is our customer base currently?
 - How do customers perform in terms of lifetime value (CLV)?

About the dataset

The dataset is made up of a single table, containing 9994 rows, each representing a single product in a transaction. The dataset can be found here¹.

The columns originally include:

1. row_id. A unique identifier for each transaction.
2. order_id. A unique identifier for each order.
3. order_date. The date when the order was placed.
4. date_key. Numerical representation of the order date.

¹ <https://www.kaggle.com/datasets/nnthanh101/aws-saas-sales/code>

5. `contact_name`. Name of the person who placed the order.
6. `Country`. Country where the order was placed.
7. `City`. City where the order was placed.
8. `Region`. Region where the order was placed.
9. `Subregion`. Subregion where the order was placed.
10. `Customer`. Name of the company that placed the order.
11. `customer_id`. A unique identifier for each customer.
12. `Industry`. Industry the customer belongs to.
13. `Segment`. Segment the customer belongs to.
14. `Product`. Name of the product that was ordered.
15. `License`. License key for each interaction.
16. `Sales`. Total sales amount for each transaction.
17. `Quantity`. Total number of items in the transaction.
18. `Discount`. The discount (%) applied to the transaction.
19. `Profit`. The profit amount of the transaction.

Limitations of the dataset

1. Because we are working with SaaS company data, it's safe to assume that the company utilizes a subscription-based business model. Unfortunately, this dataset includes only the total value of the transaction, not the value of the month-by-month cash flows.
2. There is no record of customers cancelling their subscriptions, which means an unrealistic 100% retention rate is to be expected.
3. No way to calculate the customer acquisition cost (CAC), therefore, there is no way to calculate CLV:CAC ratio.

Solutions

1. A **monthly recurring revenue (MRR) bridge** query, where monthly cash flows for each customer, based on monthly payment values are calculated. Additionally, I choose to assume that each **subscription lasts 1 year** and engineer a `subscription_end` feature.
2. Calculate **net revenue retention (NRR)** metric, which takes into account losses not only because of churn (which is expected to be 0%) but also because of revenue contraction.

3. Keeping in mind that **3:1 CLV:CAC ratio** is considered a standard benchmark for businesses, we can use it to understand how much it's worth spending to acquire additional customers.²

Data cleaning and preprocessing

This part of the analysis was done using Python and can be found among the files, uploaded to Github.

Data cleaning addressed:

1. Missing data
2. Duplicate data
3. Standardization of naming conventions
4. Data types of the columns

Additionally, I've engineered 3 features, based on the assumption that each subscription lasts 12 months, with payments occurring monthly:

1. subscription_end
2. monthly_payment
3. monthly_profit

Initial data exploration

During the 4-year period, our company provided services to 99 customers, all of whom appear to be large enterprises. These customers made purchases from all over the world – 262 cities, 48 countries, 12 subregions and 3 regions are included in the dataset. In this analysis, I've decided to treat a customer as a single unit, regardless of the order's location.

Similarly, data about customers belonging to segments was randomly generated, but after exploring the data I've chosen to remove segments column from the scope of my analysis, because it's inaccurate to treat an order made by, for example, BNP Paribas as belonging to a "Small and Medium Business" segment.

After exploring the data I've decided to test 2 assumptions:

1. H1. Effect of discount on profitability.
 - Profit loss is closely related to the size of the discount.

² David Skok's blog: <https://www.forentrepreneurs.com/ltv-cac/>

- Average discount for unprofitable deals should be significantly larger than for profitable deals.
2. H2. Effect of discount on sales seasonality.
- There will be a visible seasonal sales trend that is driven by the size of the discount.

Results:

H1. Profit-loss appears to be connected with the discount size, with average discount for unprofitable deals being significantly larger than on profitable deals.

```
---H1. EFFECT OF DISCOUNT ON PROFITABILITY---  
  
Average discount on unprofitable deals: 46.9%  
Average discount on profitable deals: 8.1%  
  
Total count of unprofitable deals: 1936  
Total count of profitable deals: 8058
```

H2. Seasonality of the sales transactions exists, but it is unrelated to the discount size, which doesn't change throughout the analysis period. Additionally, profitability of the deals follow the exact same seasonal pattern as total sales.

	sales		profit	discount
	count	sum	sum	mean
order_quarter				
2020Q1	282	74447.80	3811.23	0.16
2020Q2	392	86538.76	11204.07	0.15
2020Q3	564	143633.21	12804.72	0.16
2020Q4	755	179627.73	21723.95	0.17
2021Q1	260	68851.74	9264.94	0.15
2021Q2	444	89124.19	12190.92	0.17
2021Q3	592	130259.58	16853.62	0.15
2021Q4	806	182297.01	23309.12	0.15
2022Q1	332	92596.42	11446.34	0.15
2022Q2	594	135370.11	16084.91	0.16
2022Q3	741	144614.43	16153.50	0.16
2022Q4	913	235892.87	38042.18	0.15
2023Q1	494	118895.62	21772.23	0.15
2023Q2	692	134023.41	17165.76	0.17
2023Q3	907	200433.17	26913.45	0.15
2023Q4	1226	280594.83	27656.08	0.16

Exploratory data analysis

Time-series analysis

During this part of the analysis I'm interested in whether during the 4-year span we observe growth in monthly revenue, monthly profit and active customers. Also, how do the results of the last year, compare with the prior years.

MRR:

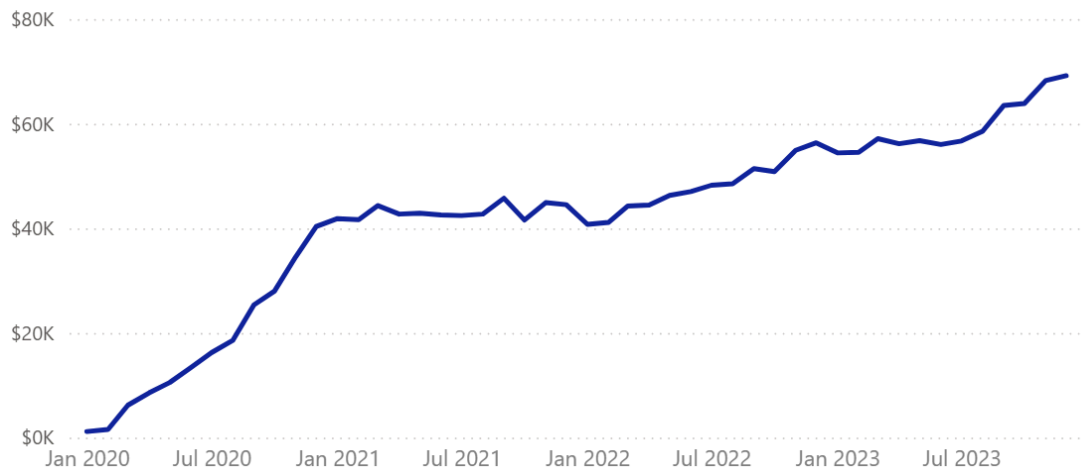
In terms of MRR (Monthly Recurring Revenue), the company has observed a **68% growth** since the end of 2020. During 2021, MRR shrunk by 3%, after which it rose by 30% during 2022 and by 25% during 2023. The company's growth pattern is encouraging as it recorded its **highest ever MRR of \$69,1k** during December of 2023.

Profit:

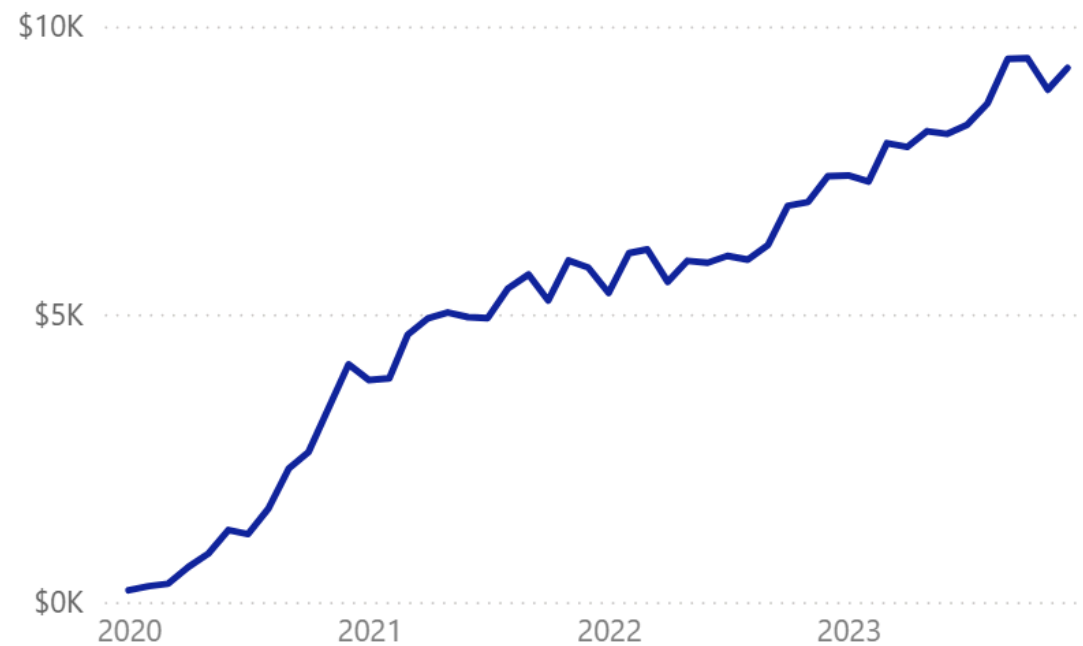
In terms of Monthly Profit, the company has observed a **142% growth** since the end of 2020. Profits were steadily rising throughout the entire analysis period, with 40% growth in 2021, 38% growth in 2022 and 25% growth in 2023. If trends continue, company will

reach \$10k monthly profit mark during next year, as it has recorded its highest ever monthly profit of \$9.4k during September of 2023.

MRR



Monthly Profit



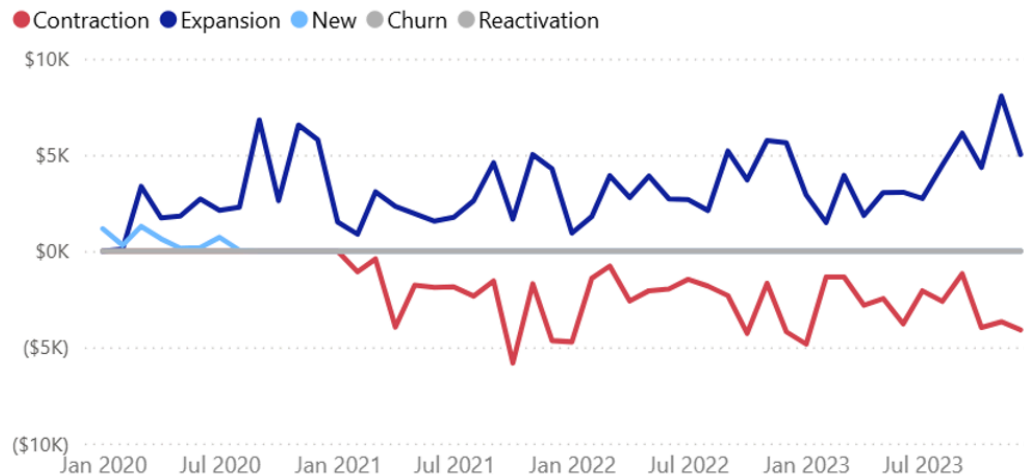
Active Customers:

Surprisingly, the company hasn't worked with any new customers for more than 3 years as the last active customer joined in September of 2020. It means that ever since that time, the company has been generating income exclusively by upselling and cross-selling different products to existing customers.

We can see from the chart below that customers tend to follow seasonal cycles, with Expansion MRR peaking during the months of September, November and December.

As of December 2023, 99 customers have an active subscription.

MRR by Type



Categorical features:

During this part of the analysis I explored each categorical feature and the effect it has on sales, profit, as well as the average discount. I'm interested if during the timespan of the analysis:

1. The categorical variable has shown better or worse performance in comparison to other variables regarding its sales volume, revenue and profit metrics.
2. The profitability or unprofitability of the categorical variable is explained by its mean discount size.

Industry:

Sales volume ranges from **2,127 transactions (21% of total)**, generated by the Finance industry, to **107 transactions (1% of total)**, generated by the Misc industry.

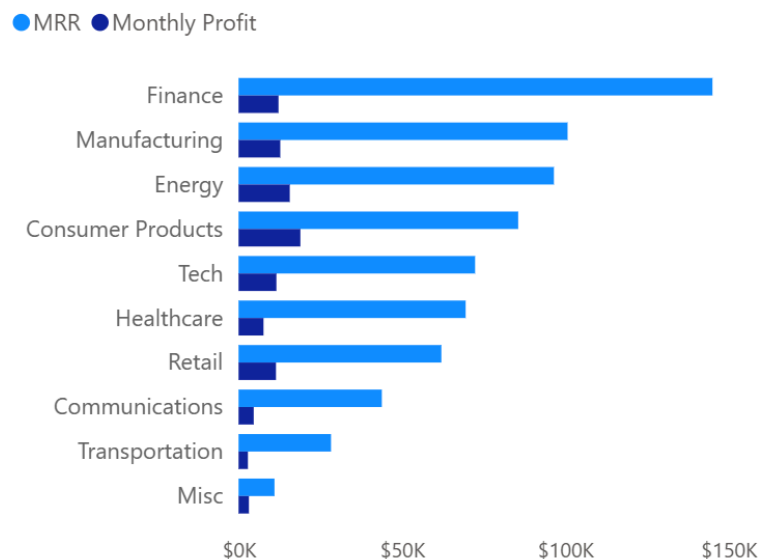
Total revenue ranges from **\$474.1k (21% of total)**, generated by the Finance industry, to **\$28.6k (1% of total)**, generated by the Misc industry. Mean revenue ranges from **\$267** by Misc industry to **\$204** by Communications industry.

Total profit ranges from **\$50.3k (18% of total)**, generated by the Finance industry, to **\$7.2k (3% of total)**, generated by the Misc industry. Mean profit ranges from **\$67** by Misc industry to **\$20** by Tech industry.

Mean discount size hovers around the dataset mean of 15%. It ranges from 16% (7 of the 10 industries), to 14% in Misc industry transactions. We can conclude that discount size does not explain the differences in profitability. Also, there were no unprofitable industries.

Overall, Finance and Energy sectors are the most important, while measured by the total profit generated. Misc, Consumer Products and Energy industries are the most important by the average profit generated.

Sales by Industry



Region:

Sales volume ranges from **4,219 transactions (42% of total)**, generated by EMEA (Europe, Middle East, Africa) region, to **2,110 transactions (21% of total)**, generated by APJ (Asia-Pacific, Japan) region.

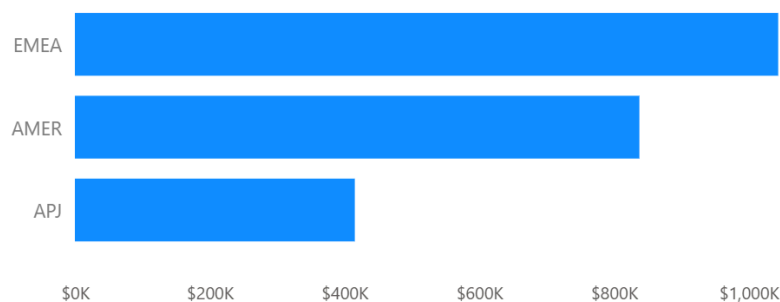
Total revenue ranges from **\$1,043.9k (45% of total)**, generated by the EMEA region, to **\$415.5k (18% of total)**, generated by the APJ region. Mean revenue ranges from **\$247** by EMEA region to **\$197** by APJ region.

Total profit ranges from **\$147.5k (51% of total)**, generated by the EMEA region, to **\$11.5k (4% of total)**, generated by the APJ region. Mean profit ranges from **\$35** by EMEA region to **\$5** by APJ region.

Mean discount ranges significantly across regions, from 11% in AMER (Americas) region to 27% in APJ region, which is almost double the size of the dataset mean of 15%. Larger overall discount seems to be the reason for low profitability numbers in Asia-Pacific and Japan region.

Overall, APJ region significantly underperformed by generating just 4% of profit despite making 21% of total transactions.

Sales Trends by Region



Subregion:

Sales volume ranges from **2,507 transactions (25% of total)**, generated by the NAMER (North America) subregion, to **110 transactions (1% of total)**, generated by the EU (Italy) subregion.

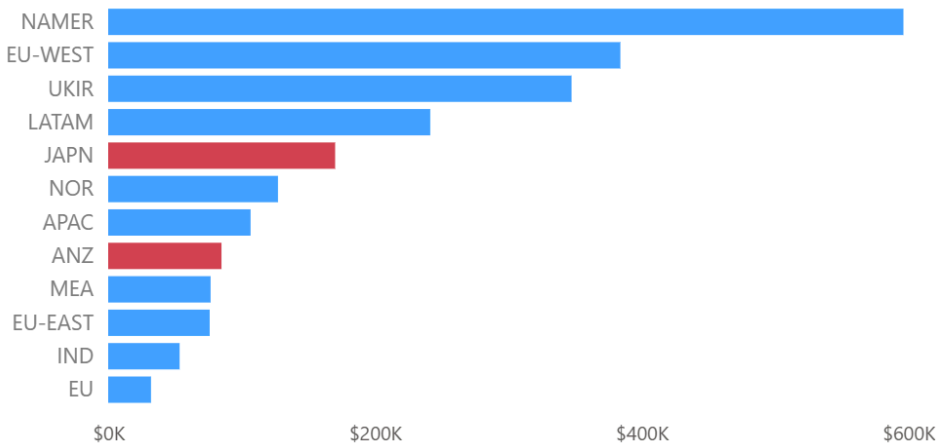
Total revenue ranges from **\$596.3k (26% of total)**, generated by the NAMER subregion, to **\$32.1k (1% of total)**, generated by the EU (Italy) subregion. Mean revenue ranges from **\$360** by IND (India) subregion to **\$161** by ANZ (Australia, New Zealand) subregion.

Total profit ranges from **\$109.8k (38% of total)**, generated by the NAMER subregion, to **-\$25.7k (-9% of total)**, generated by the JAPN (Japan) subregion. Mean profit ranges from **\$123** by IND subregion to **-\$26** by JAPN subregion.

Mean discount ranges from 0% in India and Italy to 37% in Japan and Australia, New Zealand subregions, which coincidentally are the only unprofitable subregions. Discount sizes explain the difference between subregions, with all the most profitable subregions having a mean discount size of 5% or less.

Sales Trends by Subregion

- Red subregions recorded negative profit



Product:

Sales volume ranges from **1,842 transactions (18% of total)**, generated by ContactMatcher, to **68 transactions (1% of total)**, generated by Alchemy.

Total revenue ranges from **\$410.4k (18% of total)**, generated by ContactMatcher, to **\$3.0k (0% of total)**, generated by Storage. Mean revenue ranges from **\$2,199** by Alchemy to **\$14** by Storage.

Total profit ranges from **\$55.6k (19% of total)**, generated by Alchemy, to **-\$3.5k (-1% of total)**, generated by Marketing Suite. Mean profit ranges from **\$818** by Alchemy to **-\$15** by Marketing Suite.

Mean discount ranges from 7% for Marketing Suite – Gold and Chatbot Plugin to 35% in Contact Matcher, although in this case, not discount, but difference in product's prices explains the difference in profitability.

Alchemy, the most profitable product both totally and overall, accounted for a 16% mean discount rate, while Marketing Suite – the only unprofitable product, accounted for a 21% mean discount rate – hardly a significant difference between the two.

Important to note that ContactMatcher – a product with the highest sales volume has the largest, 35% mean discount and barely a \$12k total profitability, over 4 years. It can be suggested that lowering the discount for this particular product could be very profitable.

Products - Total Sales

Product Name	Total Purchases	Total Sales	Total Profit	Discount (%)
Alchemy	68	\$149,528	\$55,618	16%
Site Analytics	889	\$330,007	\$44,516	15%
Data Smasher	775	\$167,380	\$41,937	8%
Support	1560	\$125,153	\$32,864	8%
FinanceHub	981	\$340,935	\$32,136	13%
Marketing Suite - Gold	846	\$223,844	\$21,279	7%
OneView	466	\$107,532	\$18,138	17%
SaaS Connector Pack	957	\$91,705	\$13,059	14%
ContactMatcher	1842	\$410,378	\$12,496	35%
SaaS Connector Pack - Gold	254	\$16,476	\$6,964	8%
ChatBot Plugin	796	\$27,119	\$6,528	7%
Big Ol Database	115	\$189,239	\$3,385	31%
Storage	217	\$3,024	\$950	8%
Marketing Suite	228	\$114,880	(\$3,473)	21%
Total	9994	\$2,297,201	\$286,397	16%

Country (top 20):

Sales volume ranges from **2,001 transactions (20% of total)**, generated by the United States, to **124 transactions (1% of total)**, generated by Belgium.

Total revenue ranges from **\$457.7k (20% of total)**, generated by the United States, to **\$17.4k (1% of total)**, generated by Belgium. Mean revenue ranges from **\$359** by India to **\$141** by Belgium.

Total profit ranges from **\$76.4k (27% of total)**, generated by the United States, to **-\$25.7k (-9% of total)**, generated by Japan. Mean profit ranges from **\$123** by India to **-\$36** by Mexico.

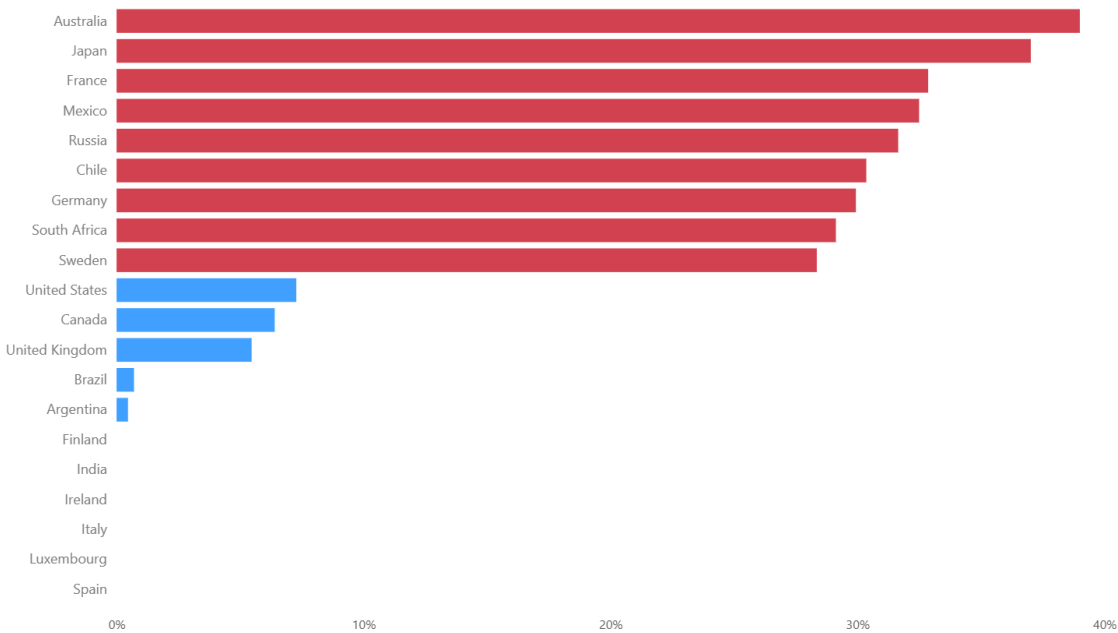
Country level analysis has shown that discounts are applied primarily at the country level, not at the company level or regional level, with profitable regions containing multiple unprofitable countries.

Unprofitable countries: Russia, Japan, France, Australia, Mexico, Sweden, Chile, South Africa, Belgium.

Discounts are the strongest explanation for a country being unprofitable, with unprofitable countries having a discount rate of 28% or larger and profitable countries having a discount rate of 20% or lower.

Average Discounts by Country

- Red countries recorded negative profit



It is also possible that nuances in taxation are included in the discount rate either directly or indirectly (selling to the customers', for example, branch in Germany, for a lower price to make it worthwhile for the customer). Because this dataset doesn't contain data that could answer this question, I'm going to assume that the discount rate is arbitrarily higher in some countries than in others.

Main analysis

Core business objective is:

- Where is **profit loss** occurring and how can we **leverage high-value customer segments** to drive further growth?

To help gain insights on the objective, the following hypothesis was raised. It was answered by the A/B test:

H1. High discounts are the main driver of profit loss, hence, capping the discount rate will bring in extra profitability.

- H0 – the number of unprofitable deals with discount ≤ 0.2 and discount > 0.2 is the same. $p_1 = p_2$.
- Ha – the number of unprofitable deals with discount ≤ 0.2 is lower than with the discount > 0.2 . $p_1 < p_2$.

- Alpha – 0.01 for a more conservative approach.

Afterwards, the following analyses were made:

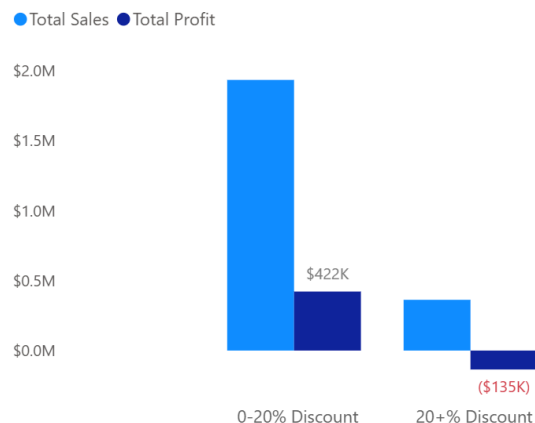
1. **Cohort analysis.** By dividing customers into cohorts, we were able to calculate NRR (net revenue retention) patterns and gather additional insights about customer behavior.
2. **CLV (Customer Lifetime Value) analysis.** Was used to answer the question which customer cohorts are high value or low value.
3. **RFM (Recency, Frequency and Monetary) analysis.** The goal is to divide customers into actionable segments according to their behavioral patterns.

A/B test. Discount's effect on profitability

Here we have found that **1361 of 1393 transactions (98%)** with a discount rate of over 20% **have been unprofitable**, while 575 out of 8601 transactions (7%) with a discount rate of 20% or lower have been unprofitable.

We have also found that despite generating \$362,7k in sales (15.7% of total), deals with a discount rate of over 20% have resulted in \$135,3k in profit loss over the 4-year period.

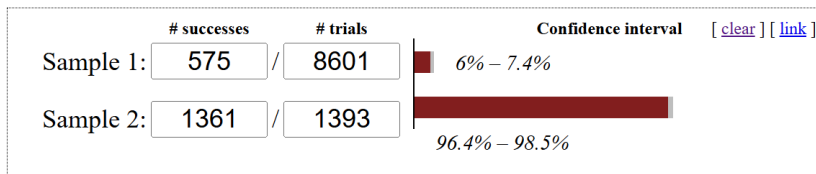
Total Sales by Discount Segment



A/B test³ has shown a clear effect of discount rates on the occurrence of unprofitable deals and therefore, we can reject the null hypothesis.

³ A/B test was done by using Evan Miller's Chi-Squared Test calculator.
<https://www.evanmiller.org/ab-testing/chi-squared.html>

Question: Does the rate of success differ across two groups?



Verdict:

Sample 2 is more successful

($p < 0.001$)

Confidence level: 99%

Results:

- We have identified a segment where most of the profit loss was occurring – high discount rate sales.

Recommendations:

- To implement a rule of 20% maximum discount on all transactions.

Cohort analysis

99 customers were grouped into monthly cohorts based on their initial purchase. There were a total of 9 monthly cohorts, all of whom were in the first year of the analysis. Additionally, as was expected, there is a 100% retention rate across all cohorts, meaning all the customers have made at least 1 purchase every 12 months.

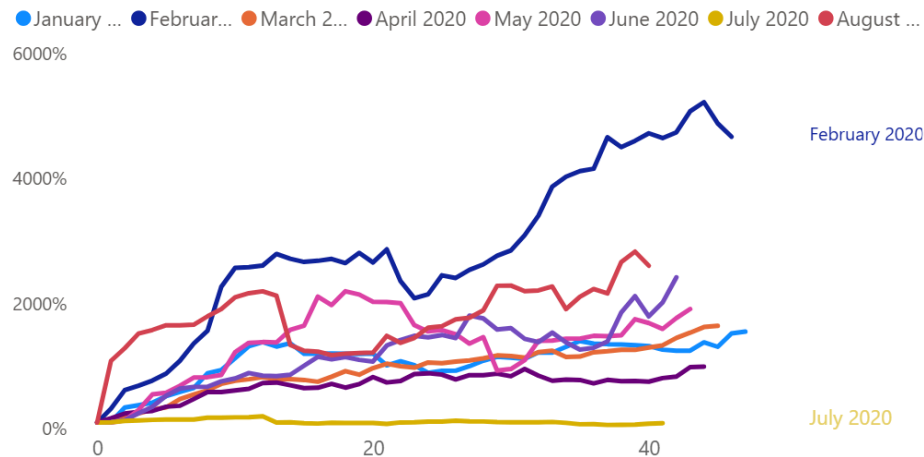
Size of the cohorts is also unequal, with 3 cohorts having less than 5 customers.

Lowest NRR (net retention rate) was recorded by the July 2020 cohort (made up of 2 customers) recording a 90% NRR in the final month of the analysis.

All the other cohorts have reached at least 900% NRR in the final month with the February 2020 cohort reaching the highest rate of 4668%.

Finally, overall NRR during the final month of the analysis (December 2023) has been at 1614%, which means that monthly, **customers spend over 16 times more** for the company's services, compared to their first month of activity.

NRR by Cohort



Results:

- The company demonstrates an exceptional ability to retain customers, as well as to find upselling and cross-selling opportunities throughout the years.
- NRR of 1614% shows an incredible track record with existing customers.
- Only problem - no new customers for over 3 years.

Recommendations:

- Try to attract new customers instead of continuing to rely on upselling and cross-selling to the existing customer base.

CLV analysis

In this analysis, I've utilized David Skok's CLV formula⁴ that was made to consider very long customer lifetimes and negative churn – the exact patterns we are encountering in this dataset.

The formula:

⁴ David Skok's blog: <https://www.forentrepreneurs.com/ltv/>

This requires us to come up with a new formula for LTV:

$$LTV = ARPA \times Gross\ Margin\% \times \left(\frac{1}{(1-K)} + \frac{G \times K}{(1-K)^2} \right)$$

Where G is the growth rate for the customers that have not churned, and K is defined by the following formula:

$$K = (1 - Churn) \times (1 - Discount\ Rate)$$

ARPA - average revenue per user

Gross margin – monthly profit / monthly revenue

G - average growth rate for retained customers (expansion revenue – contraction revenue). In this analysis I've calculated the average growth rate as the mean average growth rate of all the preceding months, to account for unpredictable monthly shifts.

(Monthly) discount rate - company's Weighted Average Cost of Capital (WACC).

According to David Skok's recommendation, a yearly discount rate of 20% for private companies was chosen⁵.

By implementing this formula, I'm able to arrive at a reasonable lifetime value statistic despite an unreasonably high 100% retention rate.

By including growth rate (which includes revenue contractions), we can calculate not only better but also worse months in terms of CLV that would be impossible with the default formula.

By applying a discount rate, we can account for possible risks of market change, new competition, changes in technology platforms, inflation and so on.

The analysis:

Across all cohorts, the Customer Lifetime Value is projected to be at ~\$42k, which is more than double of Cumulative ARPU (Average Revenue per User) that the customers have generated over the 4-year period.

⁵ According to D. Skok the earlier the stage of the startup and the greater the risk in the business, the higher the discount should be. In this case, I've chosen to assume that the company is new.

Furthermore, the actual CLV in the final month of cohort's activity, without counting projections, have exceeded \$70k in 3 of the cohorts. This indicates exceptional performance that the company has had with its active customer base.

Monthly Cohorts	0	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
January 2020	453	46,622	43,444	52,255	52,229	55,997	66,103	65,491	63,818	58,010	55,908	53,669	51,806	53,405	50,539	54,431	42,848	54,655	53,878
February 2020	193	24,429	27,713	52,723	30,940	35,525	46,913	45,312	72,810	77,679	90,189	94,581	89,157	98,318	99,167	95,738	63,679	79,658	
March 2020	383	44,744	47,349	51,416	54,513	47,726	46,031	50,398	55,982	53,824	53,740	51,714	51,492	60,422	71,774	72,269	67,891		
April 2020	692	54,264	61,601	55,399	48,351	47,726	45,143	23,780	30,353	26,159	29,305	29,250	31,025	33,188	20,680	21,758			
May 2020	-43	25,068	19,101	46,287	44,904	43,995	43,098	42,547	39,511	38,957	43,247	45,371	37,790	43,519	42,198				
June 2020	200	32,718	22,208	19,528	26,707	23,841	22,533	25,497	25,751	51,744	58,781	58,680	51,559	76,130					
July 2020	1,117	11,817	11,447	11,460	15,104	11,694	9,009	18,423	14,635	15,904	22,515	25,027	26,299						
August 2020	118	113,182	107,087	101,691	99,877	85,745	107,210	107,344	95,654	99,686	98,324	87,276							

On the other hand, at the end of the 2023 calendar year, CLV went down slightly in all cohorts and the projected value for last month's CLV growth was -17%, which might indicate that the CLV growth has largely stopped, among the current customer base.

If we take the 3:1 CLV:CAC ratio to be the baseline of what it means for a new customer acquisition to be profitable, we expect to spend no more than \$14k per new customer. A more conservative approach would be to set a benchmark based on the expected payback period being no longer than 12 months⁶ which would be at about \$8.4k, if we used last month's ARPU.

Results:

- CLV projected at \$42k is further proof that the company is ready for expanding its customer base.
- CLV growth last month: -17%. Potential sign that CLV growth for existing cohorts has stabilized.
- Acceptable CAC: Between \$8.4k and \$14k.

Recommendations:

- Invest in new customer acquisition.

RFM analysis

Because of the dataset containing just 99 customers, I've decided against using the classic 1-5 segmentation model, as there would be more segments than customers. Instead, the data was segmented in the following fashion.

⁶ David Skok's blog

https://www.forentrepreneurs.com/saas-metrics/?_ga=2.21488443.1622165577.1712153052-184887378.1712153052

- R (recency) – 84/99 customers have purchased something in the final month of the analysis while the others haven't. Because of this, I've divided the data into 2 buckets of unequal size, based on whether they have purchased anything in the last month.
- F (frequency) – total number of distinct orders that were placed. This metric was divided into 3 buckets of equal size, with possible values ranging from 13 to 103.
- M (monetary) – total sales revenue, weighed by discount rates ($\text{sales} * (1 - \text{discount})$). By applying such a formula we make sure the customers that bought more using larger discounts will get a lower monetary score. Possible monetary values range from \$5k to \$38k.

RFM segments:

High value customers: Champions ($R = 2, F = 3, M = 3$) and Loyal Customers ($R = 2, F \geq 2, M \geq 2$). These 2 segments represent customers that contribute significantly to overall revenue and are made up of loyal clients who are more likely to make repeat purchases and engage more with our company. By implementing targeted marketing strategies, businesses can nurture relationships with these clients and foster long-term loyalty.

Mid value customers: Potential Loyalists ($R = 2, F \leq 2, M = \text{any}$). This segment represents customers that have an average or lower level of engagement with a high value of recency and a varied monetary value. They are supposed to represent customers who are neither high performers, nor at risk but could be offered incentives to transition into loyal customers.

Low value customers: Recent Customers ($R = 2, F = 1, M = 1$) and Hibernating ($R = 1, F = 1, M = 1$). These segments represent customers who are at the lower end of engagement currently but could be offered incentives to engage more with our company. Important to keep in mind that because of the high overall recency of this dataset, even Hibernating customers can't be considered truly lost and retain the potential of reintegration.

Customers at risk: At Risk ($R = 1, F \geq 2$), Can't Lose Them ($R = 1, M = 3$). These segments represent customers that are valuable in terms of frequency of monetary metric but are having relatively low recency. It's imperative to drive engagement with these customers through surveys, feedback forms or direct discussions. Implementing targeted communication strategies can reignite their interest in our company and encourage renewed purchasing activity.

“Needs Attention” contains customers that make up a peculiar combination of RFM categories that wasn’t expected by the analyst and therefore, requiring attention to assess the unique situation.

In this analysis 22 customers have been classified as “Champions”, 33 as “Loyal Customers”, 11 as “Potential Loyalists”, 2 as “At Risk”, 2 as “Can’t Lose Them”, 11 as “Hibernating”, 17 as “Recent Customers” and 1 as “Needs Attention”.

Customer Segments

Customer Segment	No. of Customers	Total MRR	Total Monthly Profit
Loyal Customers	33	\$675,722	\$89,710
Champions	22	\$698,318	\$80,519
Potential Loyalists	11	\$186,191	\$26,802
Recent Customers	17	\$191,759	\$17,506
Hibernating	11	\$132,076	\$16,875
Cant Lose Them	2	\$71,101	\$14,221
At Risk	2	\$40,428	\$6,552
Needs Attention	1	\$16,583	\$2,148
Total	99	\$2,012,179	\$254,332

Results:

- 55 customers are champions/loyal customers, which means that the company has successfully created a loyal customer base that is likely to engage in the future. This means a stable base for future growth as upselling is still a possibility.
- 4 customers have been assigned to “At Risk” and “Can’t Lose Them” segments which means there is a small and manageable customer base that require extra attention via targeted communication.

Recommendations:

- The company should provide personalized suggestions, targeted marketing strategies and investigate upselling (or cross-selling) strategies to retain the high engagement of the high value customer base.
- The company should give special consideration to 4 customers at “At Risk” and “Can’t Lose Them” segments, to find a reason for their relatively low recency metric.

Conclusions

Over the 4-year analysis period, the company demonstrated impressive financial growth, achieving a 68% increase in MRR and a 142% increase in monthly profits by the end of 2023.

However, uncalibrated discounting emerged as the primary driver of profit loss, with an A/B test proving that 98% of transactions with discounts over 20% resulted in a total loss of \$135.3k. This margin erosion is highly visible in the Asia-Pacific region, where subregions like Japan and Australia became entirely unprofitable due to extreme 37% average discount rates.

On a product level, Marketing Suite remains the only unprofitable product line in the dataset, and since its 21% discount rate does not heavily diverge from highly profitable lines like Alchemy, its losses are likely tied to other factors that require further investigation.

On the customer side, the company boasts an exceptional 100% customer retention rate across all historical cohorts, culminating in an extraordinary overall Net Revenue Retention of 1614% by the final month. While this demonstrates an incredible ability to upsell and cross-sell existing accounts, it highlights a severe growth stagnation since no net-new customers have been acquired for more than three years.

Customer Lifetime Value metrics further reflect a strong underlying business model, with the average CLV projected at \$42k, which is more than double the historical cumulative ARPU generated by these accounts. This massive financial runway indicates that the company is fully prepared to fund new customer acquisition campaigns using an acceptable acquisition cost benchmark between \$8.4k and \$14k.

Furthermore, behavioral RFM analysis confirms that 55 out of the 99 active enterprise clients are clustered as high-value Champions or Loyal Customers, providing a highly reliable foundation for future recurring revenue. With only 4 total accounts flagged as being actively at risk or valuable clients with low recency, the company maintains a highly secure, satisfied, and predictable core enterprise base.

Recommendations

1. Establish Strict Discount Caps
 - Cap standard sales discounts at a hard maximum of 20%.

- Eliminate regional discounting practices.
- 2. Search for new customers to break the 3-year stagnation
 - Actively expand the user base by launching marketing campaigns to attract fresh enterprise clients.
 - Customer acquisition budget should be between \$8.4k and \$14k per new account.
- 3. Prioritize high value customer segments
 - Personalized marketing strategies to investigate further upselling, cross-selling opportunities for 55 Champions/Loyal Customers.
 - Send customer surveys to 4 customers in “At Risk” and “Can’t Lose Them” segments.
- 4. Investigate the reason for Marketing Suite’s unprofitability
 - Organize an A/B test to determine if it’s worth to continue selling “Marketing Suite” product.
 - Compare it against “Marketing Suite - Gold”.